

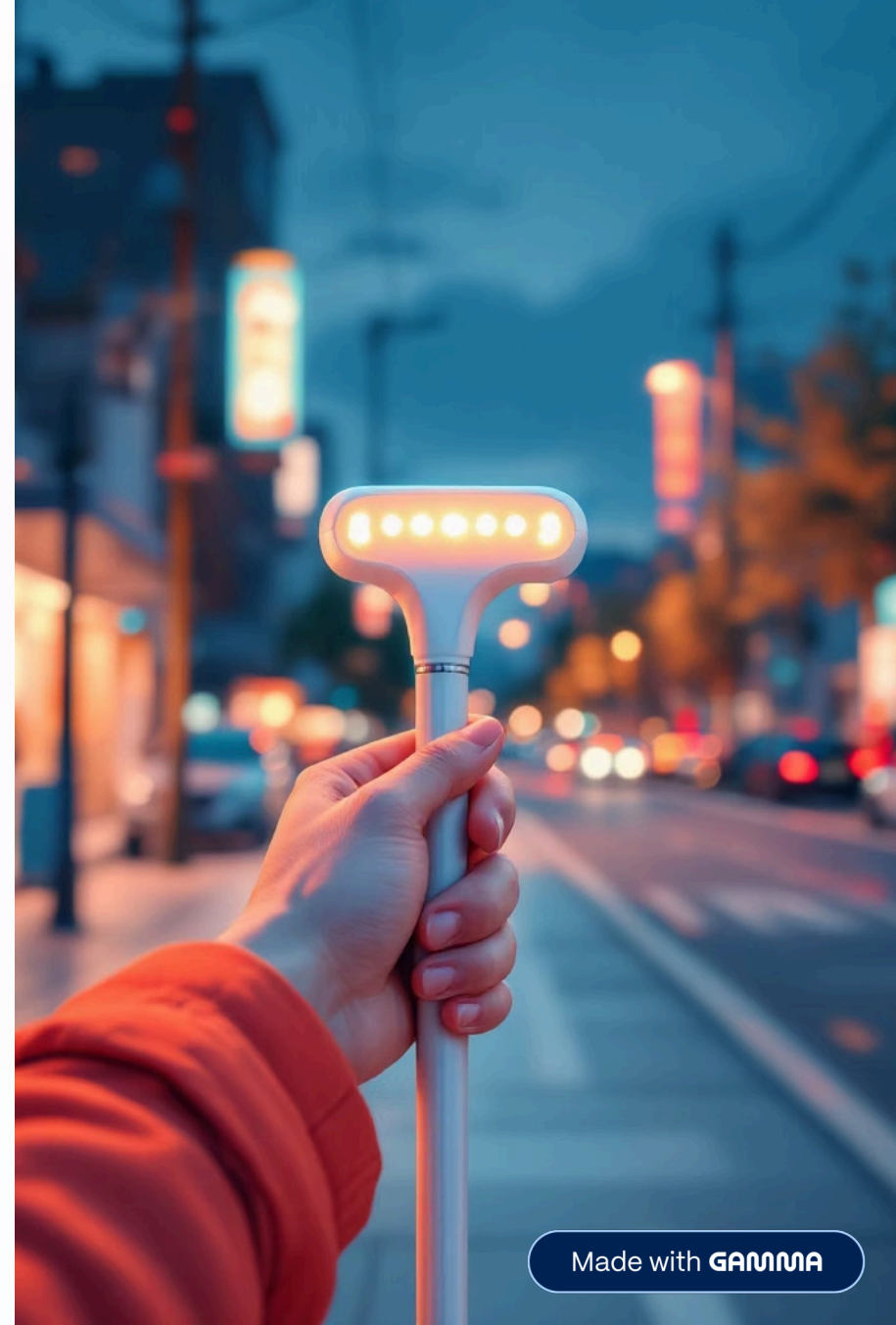
Smart White Cane with AI Navigation

Helping visually impaired users navigate safely and independently

Group 2

PH3042 - Robotics & Automation

University of Colombo



The Problem

Limited coverage

Traditional canes detect near-ground obstacles only — hazards above waist level remain unseen.

No guidance

Most canes provide no real-time routing or contextual guidance for complex environments.

Accessibility gaps

Existing solutions lack intuitive feedback patterns and easy smartphone integration for everyday use.





Our Solution

The Smart White Cane integrates sensing, embedded processing, AI vision and accessible mobile software to provide continuous obstacle awareness and route guidance.



360° Obstacle Detection

Ultrasonic sensors detect obstacles above ground level and at cane tip.



Directional Haptics

Three haptic motors encode direction and distance via distinct vibration patterns.



AI Navigation

ESP32-CAM + embedded ML for indoor markers; smartphone handles outdoor routing and voice guidance.

System Architecture

Overview of components and interactions — distributed processing with embedded vision and mobile GPS.

ESP32-S3

Central controller: sensor fusion, real-time inference, decision logic, Wi-Fi comms.

Ultrasonic Sensor

Distance measurement for near-field obstacle detection (20–200 cm).

ESP32-CAM

Indoor visual marker detection using compact ML models on-device.

Mobile App

Outdoor GPS routing, voice guidance, and system UI; receives status via Wi-Fi.

Outputs

Directional haptic motors (3) and speaker for audio prompts.

System Workflow — Data Flow

Stepwise operation for each control loop iteration (real-time loop).

1. Sensors collect data: ultrasonic (range), ESP32-CAM (marker frames).
2. ESP32-S3 performs preprocessing and sensor fusion.
3. Decision algorithm computes obstacle position & navigation instruction.
4. Haptic motors deliver directional vibration feedback to user.
5. Mobile app provides GPS voice navigation and route updates.
6. Safety priority: obstacle alerts override navigation commands immediately.

Core System Workflow



The pipeline fuses proximity sensing and vision. Local decisions (haptics) are fast; the mobile app provides route planning and high-level voice instructions via Wi-Fi.

Haptic Obstacle Detection

Three independent haptic motors around the handle convey obstacle direction and urgency:

Left motor → obstacle on user's left

Bottom motor → obstacle at cane level or fall risk

Right motor → obstacle on user's right

Distance → vibration intensity;
proximity → higher frequency pulses. Pattern vocabulary is short, distinct, and learnable.



Directional Cues

Direction encoded by motor location.



Distance Encoding

Stronger pulses signal closer obstacles.



Intuitive Learning

Short training enables rapid pattern recognition.

AI Navigation Modes

Outdoor Navigation

Smartphone uses Google Maps for routing.

App translates turns into voice prompts and directional cues to the cane over Wi-Fi.

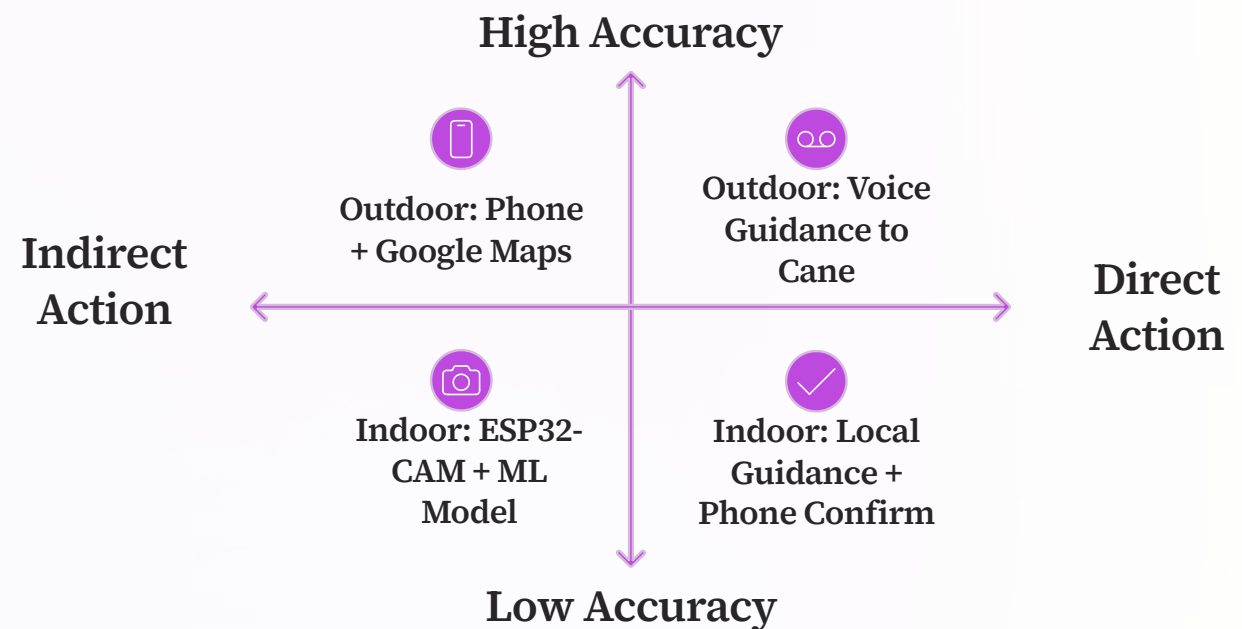
Real-time obstacle alerts override route prompts for safety.

Indoor Navigation

ESP32-CAM captures frames; embedded ML model recognizes floor/wall signs.

Markers (QR-like) indicate room IDs, turns, and waypoints.

Local guidance via haptics + voice confirmations from the phone.



Mode switching is automatic based on GPS availability and sign detection confidence.

Mobile Application — Accessibility First

Accessible UI

High-contrast, large touch targets, screen-reader optimized. Built-in voice assistant reads elements aloud.

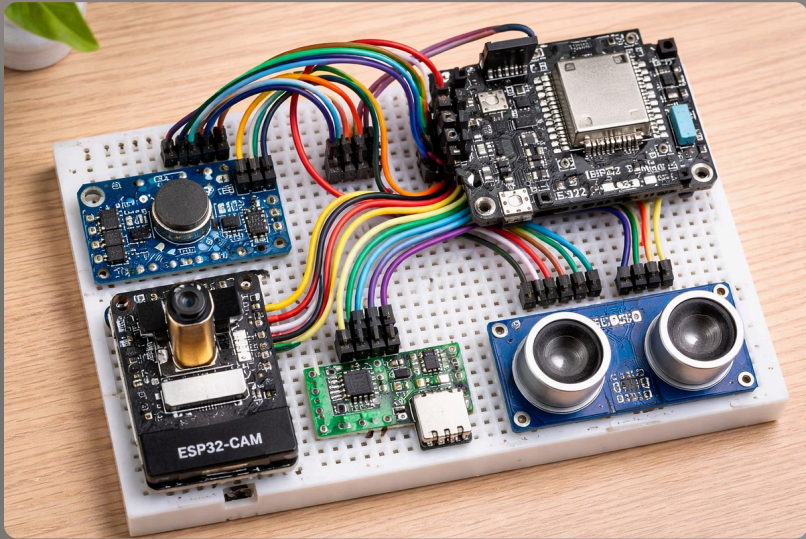
Simple Controls

Double-tap interface and long-press gestures for mode selection and emergency stop.

Two Main Modes

Indoor Navigation — marker-based guidance. Outdoor Navigation — map routing with live obstacle alerts.

Hardware Architecture



ESP32-S3

Main controller — sensor fusion, haptic control, Wi-Fi comms, low-latency decision loop.

Ultrasonic Sensors

Multi-angle ranging for obstacle detection above and at ground level.

ESP32-CAM

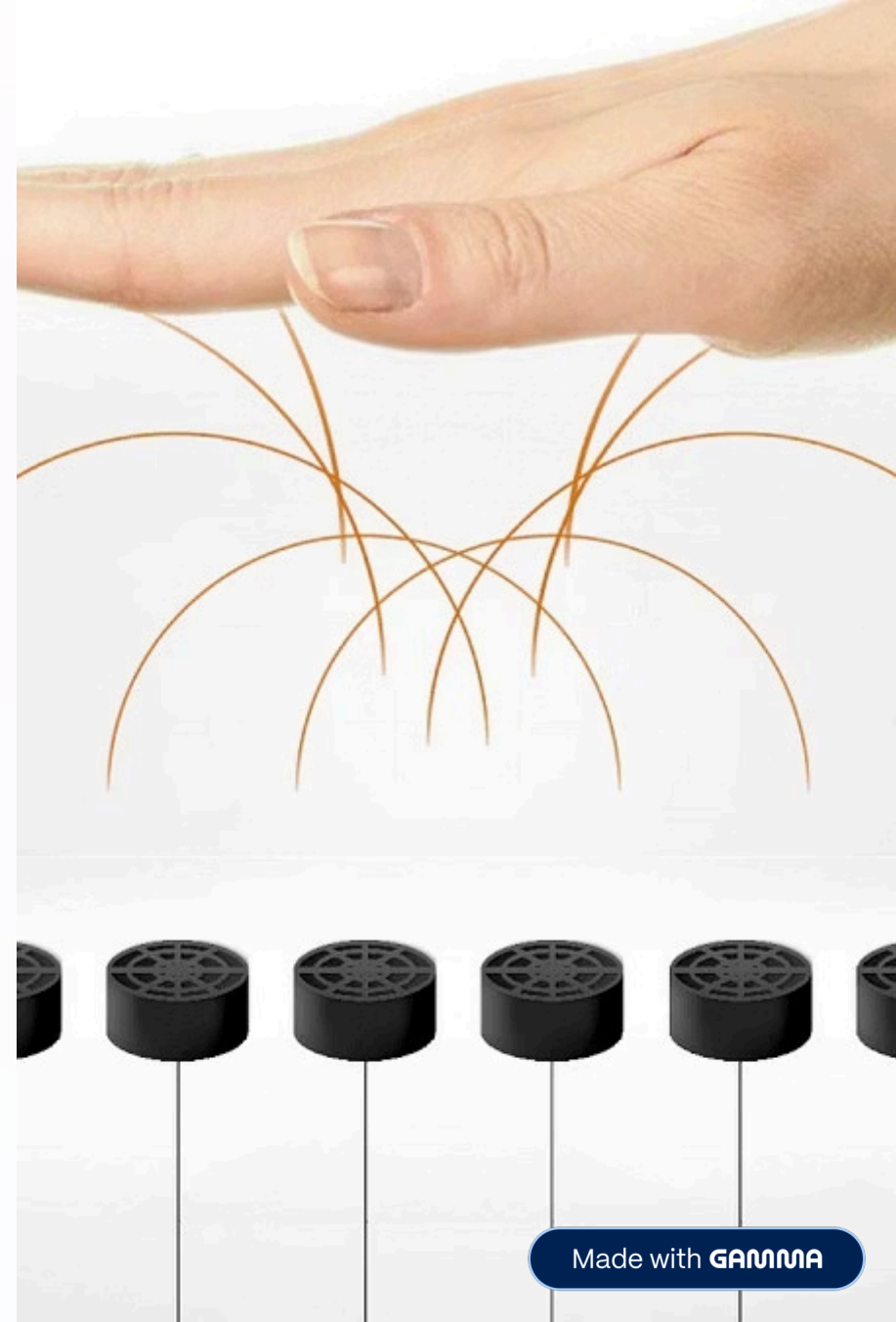
Visual marker detection using a compact ML model for indoor navigation.

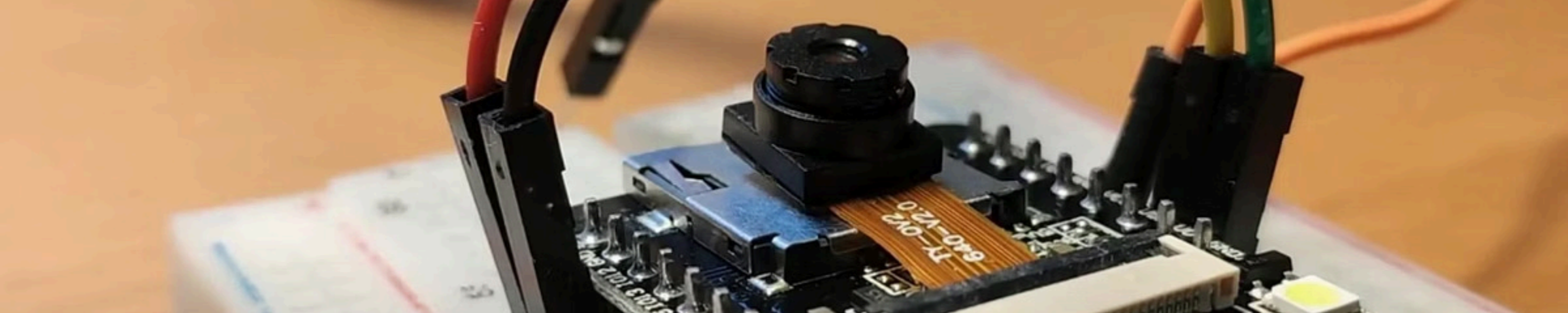
Actuators & Audio

Three haptic motors + speaker for voice prompts and confirmations.

Testing & Results

- Obstacle detection range: ~20 cm to 200 cm.
- Detection accuracy: high reliability indoors with flat/reflective surfaces.
- Haptic response time: near real-time — <1 s from detection to vibration.
- Indoor marker detection: accurate under adequate lighting; dependent on marker size and placement.
- Outdoor navigation: effective GPS route guidance via smartphone link.
- Outcome: measurable improvement in user awareness and collision avoidance.





Limitations

- Indoor navigation requires predefined visual markers for robust localization.
- Ultrasonic sensors less reliable on soft, angled, or absorptive surfaces.
- Camera performance degraded under low or uneven lighting.
- Battery life constrained by continuous sensing and Wi-Fi; impacts duty cycle.
- Outdoor routing depends on smartphone availability and GPS accuracy.

Future Improvements

Advanced Perception

On-device object recognition with lightweight deep models (quantized CNNs).

Cloud-Assisted AI

Optional cloud processing for heavier inference and model updates.

Improved Localization

SLAM or BLE beacon fusion for markerless indoor navigation.

Ergonomics & Power

Smaller form factor, weight reduction, energy optimizations for longer runtime.



Power System

Two 18650 lithium cells in parallel — nominal 3.7 V – 4.2 V

Boost converter delivers stable 5 V to modules

Integrated micro-USB charging port and hardware battery protection

Conclusion

The Smart White Cane unites robotics sensing, embedded ML vision, and accessible mobile interfaces to enhance independence and safety for visually impaired users.

Prototype demonstrates reliable haptic vocabulary, accurate indoor marker recognition, and smooth phone-driven outdoor routing. Next steps: field trials, ergonomic refinement, and model optimization for lower power.

Market Potential & Global Impact

Target Users • Visually impaired individuals worldwide • Elderly people with mobility challenges • Rehabilitation centers and assistive technology programs	Market Need • Over 285 million visually impaired people globally • Increasing demand for affordable smart assistive devices • Limited availability of low-cost intelligent navigation solutions	Competitive Advantage • Hybrid navigation (indoor AI + outdoor GPS) • Directional haptic feedback (more intuitive than audio-only) • Low-cost and scalable design • Seamless smartphone integration
Deployment Strategy • Collaborate with healthcare organizations and NGOs • Distribution through hospitals and assistive technology providers • Mobile application via app stores • User training and onboarding support	Scalability • Suitable for mass production using low-cost components • Software updates for continuous improvement • Adaptable for different countries and environments	



Thank You.....